

WJEC GCE Unit 5

Condense Notes

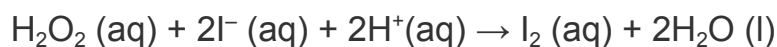
2026 Version

IGC HK Exam

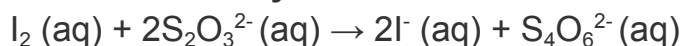


Kinetics – Iodine Clock

Reaction Equation



Sodium thiosulfate act as a delay



Overall Rate Equation

$$\text{Rate} = k [\text{S}_2\text{O}_8^{2-}] [\text{I}^-]$$

Questions

Use of H_2O – To allow the total volume to be same so volume of $\text{S}_2\text{O}_8^{2-}$ or I^- is proportional to their concentration + Fair Test

Colour change – Starch: Colourless to Blue-black

Calculate concentration of IV – $c_1 v_1 = c_2 v_2$ so eg I^- is IV $c_{\text{I}^-} v_{\text{I}^-} = c_{\text{IV}} v_{\text{Total}}$

More Objective Result – Colorimeter

More Reliable Result – Repeat experiment and take mean + Increasing range of concentrations

More Accurate Result – Higher resolution stopwatch

Use small volume of $\text{S}_2\text{O}_3^{2-}$ – Such that the change in concentration of reactant is insignificant and the time need to change the colour of the solution to dark blue can reflect the initial rate of reaction

Rate Equation Calculation

Rate Increase Factor = [Zeroth Order]⁰ [First Order] [Second Order]²

$$\text{Ex: } x_4 = (x_2)^1 (x_2)^n n=1$$

Conclusion sentence: The reaction is nth order with respect to [conc].

Arrhenius Equation

$$k = Ae^{-\frac{E_a}{RT}}$$

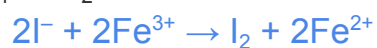
$$\ln k = \ln A - \frac{E_a}{RT} \times \frac{1}{T}$$

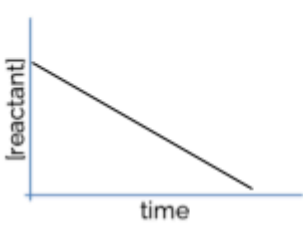
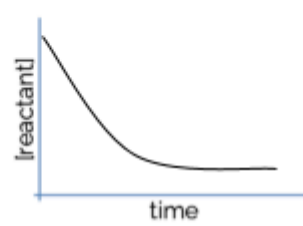
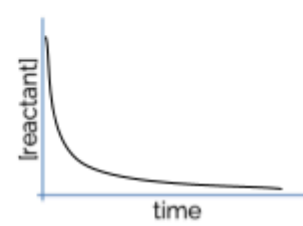
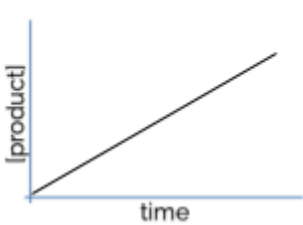
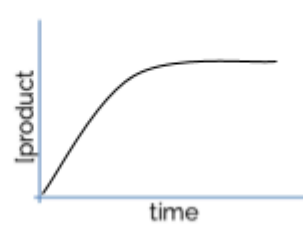
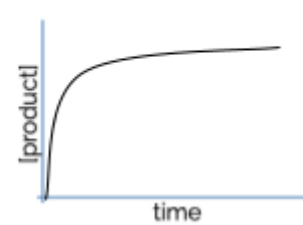
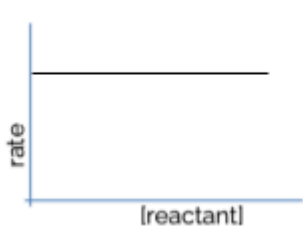
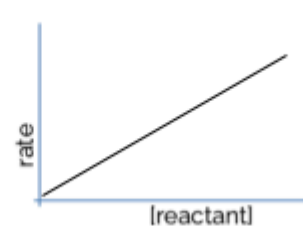
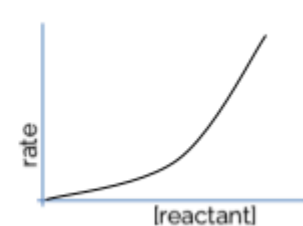
Mark Scheme

- Total volume (H_2O_2 / I^-) and H_2O must be same
- Table + Graph must include units

Disappearing Cross – $\text{S}_2\text{O}_3^{2-} + 2\text{H}^+ \rightarrow \text{SO}_2 + \text{S} + \text{H}_2\text{O}$

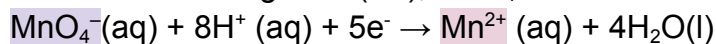
Iodine Clock Ver 2



Order of reaction	zeroth	first	second
[reactant] v time			
[product] v time			
rate v [reactant]			

Feasibility of Redox

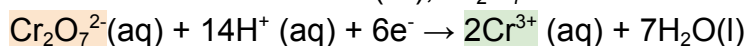
Manganate(VII), MnO_4^-



Purple $\rightarrow$ Pale Pink

MnO_2 – dark brown ppt

Dichromate(VI), $\text{Cr}_2\text{O}_7^{2-}$



Orange $\rightarrow$ Green

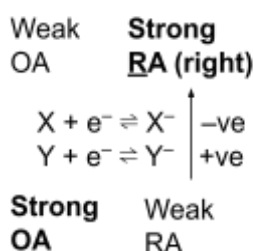
Salt Bridge

Function: Completes the circuit, allow a current to flow and keep equipotential

Substance: Filter paper soaked in saturated potassium nitrate solution

Standard Conditions
Temperature: 298K
Concentration: 1 mol dm⁻³ of HCl
Pressure: 1 atm

Platinum electrode: Inert, Conductor, Catalyst

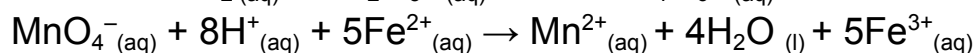
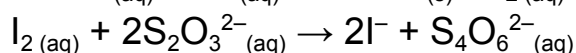
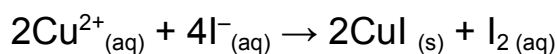


$$E^\theta_{\text{cell}} = E_{\text{reduction}} - E_{\text{oxidation}}$$

Half Equation



Memorise Equation



Compare Feasibility

Electrode reaction	$E^\ominus / \text{V}$
$2 \text{H}^+(\text{aq}) + 2 \text{e}^- \rightarrow \text{H}_2(\text{g})$	0.00
$\text{Cu}^{2+}(\text{aq}) + 2 \text{e}^- \rightarrow \text{Cu}(\text{s})$	+0.34
$\text{NO}_3^-(\text{aq}) + 4 \text{H}^+(\text{aq}) + 3 \text{e}^- \rightarrow \text{NO}(\text{g}) + 2 \text{H}_2\text{O}(\text{l})$	+0.96

Explain why copper does not react with most acids but does react with nitric acid

- Hydrogen electrode is less positive than the Copper electrode so H^+ cannot oxidise Cu to Cu^{2+}
- Nitrate electrode is more positive than copper electrode so NO_3^- can oxidise Cu to Cu^{2+}

Explain Feasibility

- Compare $E^\ominus_{\text{cell}}$ +ve / -ve
- As standard electrode potential for hydrogen peroxide is more positive than that for chromium/dichromate

- (c) Blue oxovanadium(IV) cations, VO^{2+} , can be oxidised to yellow oxovanadium(V) ions, VO_2^+ , by bromate(V) ions, BrO_3^- .

	Standard electrode potential, $E^\ominus / \text{V}$
$\text{VO}_2^+ + 2\text{H}^+ + \text{e}^- \rightleftharpoons \text{VO}^{2+} + \text{H}_2\text{O}$	+1.00
$\text{Br}_2 + 2\text{e}^- \rightleftharpoons 2\text{Br}^-$	+1.09
$\text{BrO}_3^- + 6\text{H}^+ + 6\text{e}^- \rightleftharpoons \text{Br}^- + 3\text{H}_2\text{O}$	+1.44

Use these standard electrode potentials to identify the main bromine-containing product formed when excess bromate(V) ions are added to a solution containing VO^{2+} ions. Give reasons for your answer. [3]

Br_2 (1) MUST GIVE REASON TO GAIN THIS MARK
 bromate is reduced to bromide / bromate is a strong enough oxidising agent to oxidise VO^{2+} as it has a more positive $E^\ominus$ value
 OR EMF = +0.44 V and this is positive so it is feasible (1)

bromate is a strong enough oxidising agent to oxidise the bromide formed in the first step to Br_2 as it has a more positive $E^\ominus$ value
 OR EMF = +0.35 V and this is positive so it is feasible (1)

Mention about more +ve / -ve SEP value

So it is a stronger OA / RA to oxidise / reduce the other cell

$+E_{\text{cell}}$ but not feasible

- High E_A
- Not standard condition

- (a) Indicate what reactions you would expect to occur when a solution containing iron(II) bromide is treated with chlorine gas. Use the following standard electrode potentials to explain your conclusions: [3]

$$E^\circ(\text{Cl}_2 | 2\text{Cl}^-) = +1.36 \text{ V};$$

$$E^\circ(\text{Fe}^{3+} | \text{Fe}^{2+}) = +0.77 \text{ V};$$

$$E^\circ(\text{Br}_2 | 2\text{Br}^-) = +1.09 \text{ V}.$$

- (b) Describe the observations which you would expect to accompany these reactions. [3]

(a)

- Cl_2 is a stronger oxidising agent than Br_2 and Fe^{3+} (higher E° value)
- Cl_2 oxidises Br^- to Br_2
- Cl_2 oxidises Fe^{2+} to Fe^{3+}

(b)

- Orange/brown colour forms (Br_2)
- Pale green solution turns yellow/brown ($\text{Fe}^{2+} \rightarrow \text{Fe}^{3+}$)
- Chlorine gas decolourises / green colour disappears

Chemical Test

Flame Test

Li ⁺	Red
Na ⁺	Yellow-orange
K ⁺	Lilic
Ca ²⁺	Brick red
Sr ²⁺	Red
Ba ²⁺	Apple green
Cu ²⁺	Blue-green

Other Test

$\text{SO}_4^{2-} \rightarrow \text{Ba}^{2+} \rightarrow \text{Heavy white ppt}$

$\text{CO}_3^{2-} \rightarrow \text{HNO}_3 \rightarrow \text{Effervescence} \rightarrow \text{Test CO}_2 \text{ gas by bubbling in lime water} \rightarrow \text{white ppt form (milky / cloudy)}$

Halide $\rightarrow$ Acidified Silver Nitrate $\rightarrow$ AgCl (white ppt), AgBr (cream ppt), AgI (yellow ppt) $\rightarrow$ Dilute Ammonia $\rightarrow$ Concentrated Ammonia

	Colour	Dilute NH ₃	Conc NH ₃
AgCl	White	Dissolves	Dissolves
AgBr	Cream	Insoluble	Dissolves
AgI	Yellow	Insoluble	Insoluble

Group II

- Sulphate **decrease** solubility down the group ($\text{BaSO}_4 \text{ (s)}$)
- Group II Carbonate all insoluble
- Hydroxide **increase** solubility down the group ($\text{Mg(OH)}_2 \text{ (s)}$)

Group VII

Strong OA	Weak RA
Cl_2	Cl^-
Br_2	Br^-
I_2	I^-
Weak OA	Strong RA

Halogen	State at room temp	colour	colour of vapour	colour of aqueous solution
F	gas	pale yellow		very pale yellow
Cl	gas	pale green		very pale green
Br	liquid	red-brown	orange	orange
I	solid	grey	purple	brown

Solubility

Soluble Salt	Insoluble Salt
All Group 1 Salts	
All NH_4^+ Salts	
All NO_3^- Salts	
Most Halide	Halide salt of Pb^{2+} , Ag^+ , Cu^+
Most Sulphate	BaSO_4 PbSO_4 SrSO_4 Ag_2SO_4 CaSO_4 (sparingly soluble)
Group 1 & NH_4 Carbonates (mentioned above)	Most Carbonates
Group 1 & NH_4 Hydroxide $\text{Ba}(\text{OH})_2$ $\text{Sr}(\text{OH})_2$ Hydroxide	Most Hydroxides

Amphoteric:

Solution forms a precipitate in addition of hydroxide, which is then able to redissolve in excess hydroxide, the compound is amphoteric.

Chromium (Cr) – transitional metal

Zinc (Zn) – d block

Aluminum (Al) – p block

Lead (Pb) – p block

Precipitation of Pb^{2+} ions

$\text{Pb}^{2+} \rightarrow \text{Pb}(\text{OH})_2$ White ppt $\rightarrow \text{Pb}(\text{OH})_4^{2-}$ Redissolve colourless

$\text{PbCl}_2 \rightarrow$ White ppt

$\text{PbI}_2 \rightarrow$ Canary Yellow ppt

Manganese	Manganate (VII) MnO_4^-	dark purple
	Manganate (IV) MnO_2	brown
	Manganate (II) Mn^{2+}	very pale pink
Chromium	Dichromate (VI) $\text{Cr}_2\text{O}_7^{2-}$	orange
	Chromate (VI) CrO_4^{2-}	yellow
	Chromium (III) Cr^{3+}	green
Iron	Iron (II) Fe^{2+}	very pale green
	Iron (III) Fe^{3+}	rust orange
Cobalt	Cobalt (II) Co^{2+}	pink
	Cobalt (III) Co^{3+}	blue
Copper	Copper (II) Cu^{2+}	sky blue
	Copper (I) Cu^+	colourless

$[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$	Sky Blue
$[\text{Cu}(\text{H}_2\text{O})_4(\text{OH})_2]_{(s)}$	Pale Blue / 'Duck Egg Blue'
$[\text{Cu}(\text{H}_2\text{O})_2(\text{NH}_3)_4]^{2+}$	Royal Blue
$[\text{CuCl}_4]^{2-}$	Yellow
$[\text{Co}(\text{H}_2\text{O})_6]^{2+}$	Pink
$[\text{CoCl}_4]^{2-}$	Blue
$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	Pale Green
$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$	Rust Orange
$[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$	Pale Pink

	Some NaOH	Excess NaOH
$[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$	grey-green ppt of $[\text{Cr}(\text{H}_2\text{O})_3(\text{OH})_3]$	ppt dissolves giving a deep green solution of $[\text{Cr}(\text{OH})_6]^{3-}$
$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	dark green ppt of $[\text{Fe}(\text{H}_2\text{O})_4(\text{OH})_2]$	ppt does not redissolve red brown colour might be seen as Fe^{2+} might be oxidised to Fe^{3+}
$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$	red-brown ppt of $[\text{Fe}(\text{H}_2\text{O})_3(\text{OH})_3]$	no further reaction
$[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$	pale blue (duck egg blue) ppt of $[\text{Cu}(\text{H}_2\text{O})_4(\text{OH})_2]$	no further reaction

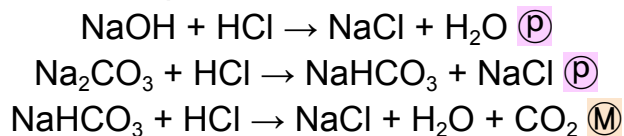
- $\text{Fe}(\text{OH})_2$ and $\text{Cr}(\text{OH})_3$ is much darker precipitate than the Fe^{2+} solution and Cr^{3+} solution
- Fe^{3+} is being oxidised from Fe^{2+} (rust iron)

Salt Preparation

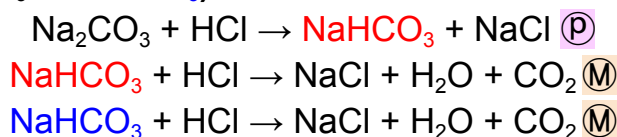
Soluble Salt Crystal	Insoluble Salt
- <u>Excess</u> Insoluble + Acid Metal Metal Oxide Metal Hydroxide Metal Carbonate - Filter off the excess insoluble solution - Heat until <u>saturated</u> - Cool down for crystal to form - Collect by filtration - Wash deionized water, dry	- Dissolve the soluble salt in deionised water - Soluble salt for ion-exchange - Collect by filtration - Wash - Dry with filter paper

Acid and Base – Double Titration

Scenario 1 (NaOH + Na₂CO₃)



Scenario 2 (Na₂CO₃ + NaHCO₃)



Risk Assessment

Toxic & Corrosive > Irritant > Flammable

NaOH – Corrosive, even if 0.1M → still very irritant + corrosive *hydrolyse protein*

Hazard	Risk	Control Variable
Toxic	Serious harm to health if ingested	Wear safety glasses, gloves and lab coats. Use small amounts, dispose of safely.
Corrosive	Could accidentally spill onto skin or into eyes during use	Wear safety glasses, gloves + lab coat throughout
Irritant	Could accidentally spill onto skin or into eyes during use	Wear safety glasses, gloves + lab coat throughout
Flammable	Could ignite and cause fire during use	No naked flames

Other Solution

- PbI₂ (canary yellow) + PbCl₂ (white)
- Fe²⁺ – pale green
- Fe³⁺ – rust orange
- I₂ (aq) orange
- I₂ add cyclohexane turn purple

Organic Chemistry

Hydrolysis

